

A Multi Task Deep Learning model for Fruit detection and Freshness Classification with GradCAM Explainability

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Abstract—This study proposes a multi-task deep learning model based on ResNet152V2 to simultaneously classify fruit type and freshness (fresh vs. rotten) from images. Leveraging the Kaggle Fresh and Rotten Fruits dataset (30,357 images across 9 fruit varieties), the model employs a ResNet152V2 backbone with a shared feature extraction trunk and dual task-specific classification branches. A smart class balancing strategy and extensive data augmentation were applied to mitigate the dataset's initial class imbalance. We benchmarked the proposed model against eight deep learning architectures (Custom CNN, VGG16, InceptionV3, EfficientNet-B4, DenseNet121, ConvNeXt-Base, Vision Transformer) and an object detection model (YOLOv3), achieving superior performance. The ResNet152V2 multi-task model attained 99.64% accuracy in fruit classification and 97.27% in freshness detection (98.45% combined accuracy), outperforming all comparators. We employ Grad-CAM visualizations to enhance interpretability, demonstrating that the model leverages structural cues (e.g., shape, stem) for fruit identification and surface cues (e.g., discoloration, spots) for freshness assessment. We further deploy the trained model as a real-time web application on Hugging Face Spaces to demonstrate its practical feasibility for instant fruit quality evaluation. These results establish a new state-of-the-art for multi-task fruit classification and advance food quality assessment and supply chain automation through accurate, interpretable, and deployable fruit freshness detection.

Index Terms—Food Quality Assessment, Multi-task CNN, Transfer learning, Grad-CAM, Real-time deployment

I. INTRODUCTION

Automatic fruit freshness detection has become increasingly critical for reducing food waste and enhancing supply chain

quality control in modern agriculture and retail industries. With global food waste reaching approximately one-third of all produced food, accurate freshness assessment systems can significantly contribute to sustainable food management. Deep learning, particularly Convolutional Neural Networks (CNNs), has emerged as the predominant solution for image-based freshness classification in recent years due to their superior feature extraction capabilities and classification performance.

The foundation for automated fruit freshness assessment was established through single-fruit studies. Ni et al. [1] pioneered transfer learning approaches for fruit freshness, where a fine-tuned GoogLeNet model achieved 98.92% accuracy in distinguishing fresh from rotten bananas, essentially matching human-level performance. Building on transfer learning approaches, Pathak and Makwana [2] developed a specialized 6-layer CNN tailored for classifying apples, bananas, and oranges as fresh or rotten, reaching 98.23% accuracy after careful hyperparameter tuning on 10,901 images. Kazi et al. [3] demonstrated the efficacy of integrating transfer learning with classical machine learning approaches. Specifically, their study employed VGG-16-derived deep features in conjunction with a quadratic SVM classifier, attaining an accuracy of approximately 98%. These findings underscore the potential of convolutional neural network features, when combined with traditional classifiers, to achieve high precision in fruit freshness classification. Comparative analysis by Nerella et al. [4] evaluated popular CNN architectures including VGG16, ResNet50, MobileNet, and EfficientNet on

common fresh/rotten fruit datasets, demonstrating that most transfer-learned CNNs exceed 95% accuracy and generally outperform models trained from scratch. The highest reported performance for single-task freshness classification came from Yuan et al. [5], who developed a hybrid CNN-BiLSTM model that achieved 97.76% accuracy by combining convolutional feature extraction with bidirectional LSTM for capturing spatial and contextual correlations in fruit images. Expanding beyond binary classification, Fahad et al. [6] tackled a three-class freshness categorization (pure-fresh, semi-fresh, rotten) using a large dataset of approximately 60,000 images covering 11 fruits and vegetables. Using VGG-16 for classification and YOLOv5 for localization, they achieved 82-84% accuracy with two different models and demonstrated a mobile application for real-time freshness detection. Multi-task learning approaches for fruit freshness classification have gained attention recently. Kang and Gwak [7] developed an ensemble of multi-task deep convolutional neural networks using transfer learning, combining ResNet-50 and ResNet-101 architectures to achieve 97.43% fruit classification and 98.50% freshness detection accuracy. Zhang et al. [8] proposed a multi-task CNN framework specifically designed for fruit freshness detection, demonstrating the effectiveness of shared feature learning for simultaneous fruit type and freshness classification, achieving 88.66% fruit classification and 93.24% freshness detection accuracy. In summary, prior studies either focus on freshness classification for a single fruit or use separate models for fruit type and freshness assessment. Only a few attempt true multi-task learning [7], [8], and those were limited in scope or required ensemble approaches with multiple models. Moreover, practical deployment with explainability is seldom addressed, with most works remaining as proof-of-concept studies. Additionally, comprehensive benchmarking of multiple state-of-the-art architectures for multi-task fruit freshness classification remains limited.

This paper addresses these gaps by providing the first benchmark of nine deep learning models for multi-task fruit classification on a diverse dataset, delivering an integrated solution with explainability features ready for real-time deployment. Our contributions include: (1) proposing a custom multi-task CNN architecture benchmarked against eight state-of-the-art models, (2) achieving superior performance of 98.45% average accuracy on a 9-fruit dataset with individual task performance, (3) providing explainability via Grad-CAM visualization, and (4) demonstrating deployment capabilities through a real-time web application.

II. METHODOLOGY

A. Dataset Description

We utilize the ‘Fresh and Rotten/Stale Fruits and Vegetables Classification’ dataset from Kaggle, uploaded by S.S. Nayak in June 2023 [9]. It contains images of 9 fruit/vegetable classes (apples, bananas, oranges, tomatoes, potatoes, cucumbers, bittergourd, okra, capsicum) in both fresh and rotten states. The original dataset had 30,357 images with significant class imbalance, ranging from 7,451 apple images to

only 684 bittergourd images. We applied a 70%/15%/15% train-validation-test split, resulting in 14,178 training samples, 3,038 validation samples, and 3,039 test samples. After smart balancing and augmentation of minority classes, the final dataset contains 20,255 samples with 9,845 fresh and 10,410 rotten images.

B. Class Balancing Strategy

To address class imbalance (ranging from 7,451 apple images to 684 bittergourd images), we applied targeted augmentation to minority classes. Classes with fewer than 800 samples (particularly bittergourd with 684 images) were augmented using transformations including rotation, width/height shifts, shear, zoom, and brightness adjustments. This balancing strategy ensured no class bias in model training while preserving the distribution characteristics of the dataset.

C. Data Preprocessing and Augmentation

All images were resized to 224×224×3 RGB format to maintain consistency with standard CNN input requirements and normalized to [0,1] scale by rescaling with factor 1./255. The comprehensive data augmentation pipeline included:

- Rotation range: $\pm 20^\circ$
- Width and height shifts: $\pm 15\%$
- Shear range: 15%
- Zoom range: 15%
- Horizontal flip: enabled
- Brightness range: [0.9, 1.1]

Class weights were computed to handle remaining imbalances, with freshness detection weights of {0: 0.973, 1: 1.029} for rotten and fresh classes respectively.

D. Training Procedure

All models were trained using Adam optimizer with initial learning rate 0.001 (reduced to 0.0005 for ViT and ResNet152V2 to stabilize training). Training used batch size of 4, optimized for Tesla P100-PCIE-16GB GPU memory constraints. Early stopping (patience=2) and ReduceLROnPlateau callbacks prevented overfitting when validation loss ceased improving. Training duration was 8-10 epochs across all models with learning rate reduction applied when validation performance plateaued.

E. Proposed ResNet-152V2 Multi-Task Architecture

The proposed ResNet-152V2 Multi-Task Model simultaneously classifies fruit type and assesses freshness within a single, end-to-end network. Fig. 1 highlights the three main blocks of the design: (i) a *ResNet-152V2 backbone* for high-quality feature extraction, (ii) a *shared dense trunk* that captures task-agnostic representations, and (iii) two *task-specific heads* for classification and freshness detection.

Backbone Architecture: A pre-trained ResNet-152V2 (ImageNet) provides the first 152 residual layers. To balance adaptation and generalisation we freeze the first 122 layers (`base_model.layers[:30]`) and fine-tune only the final 30, allowing domain-specific refinement while preserving low-level features such as edges and textures.

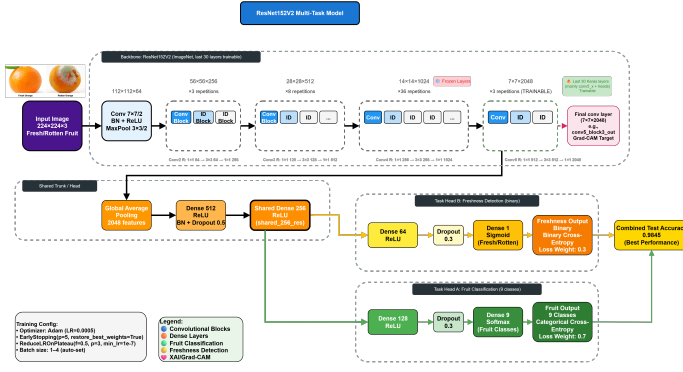


Fig. 1. Architecture of the proposed multi-task ResNet-152V2 model showing backbone, shared trunk, and dual task-specific branches.

Global Feature Aggregation: The backbone's $7 \times 7 \times 2048$ activation tensor is compressed with Global Average Pooling (GAP) to a 1×2048 vector,

$$\text{GAP}(f) = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W f(i, j),$$

which removes spatial redundancy and yields translation-invariant descriptors.

Shared Dense Trunk: GAP features pass through a 512-neuron Dense layer with ReLU, Batch Normalisation, and 0.5 Dropout, followed by a 256-neuron *shared* Dense layer:

$$f_{\text{shared}} = \text{ReLU}(\mathbf{W}_{\text{shared}} f_1^{\text{drop}} + \mathbf{b}_{\text{shared}}).$$

This layer learns joint cues—colour constellations, surface patterns, and texture gradients—useful for both downstream tasks.

Multi-Task Output Branches: From f_{shared} the network diverges into two lightweight heads:

- 1) Fruit Classification Branch (9 classes)

$$f_{\text{fruit}} = \text{ReLU}(\mathbf{W}_{\text{fruit}} f_{\text{shared}} + \mathbf{b}_{\text{fruit}}), \quad (1)$$

$$f_{\text{fruit}}^{\text{drop}} = \text{Dropout}(f_{\text{fruit}}, 0.3), \quad (2)$$

$$P_{\text{fruit}} = \text{Softmax}(\mathbf{W}_{\text{out,fruit}} f_{\text{fruit}}^{\text{drop}} + \mathbf{b}_{\text{out,fruit}}). \quad (3)$$

A 128-neuron dense layer precedes the 9-way softmax, giving the head enough capacity to model subtle inter-class differences (e.g., orange vs. mandarin) without bloating the parameter count. The softmax probabilities support confidence-aware decisions and enable categorical cross-entropy optimisation.

- 2) Freshness Detection Branch (binary)

$$f_{\text{fresh}} = \text{ReLU}(\mathbf{W}_{\text{fresh}} f_{\text{shared}} + \mathbf{b}_{\text{fresh}}), \quad (4)$$

$$f_{\text{fresh}}^{\text{drop}} = \text{Dropout}(f_{\text{fresh}}, 0.3), \quad (5)$$

$$P_{\text{fresh}} = \text{Sigmoid}(\mathbf{W}_{\text{out,fresh}} f_{\text{fresh}}^{\text{drop}} + \mathbf{b}_{\text{out,fresh}}). \quad (6)$$

Here a 64-neuron dense layer sharpens sensitivity to micro-texture cues such as mould spots, colour dulling, and surface wrinkling. The final sigmoid outputs a

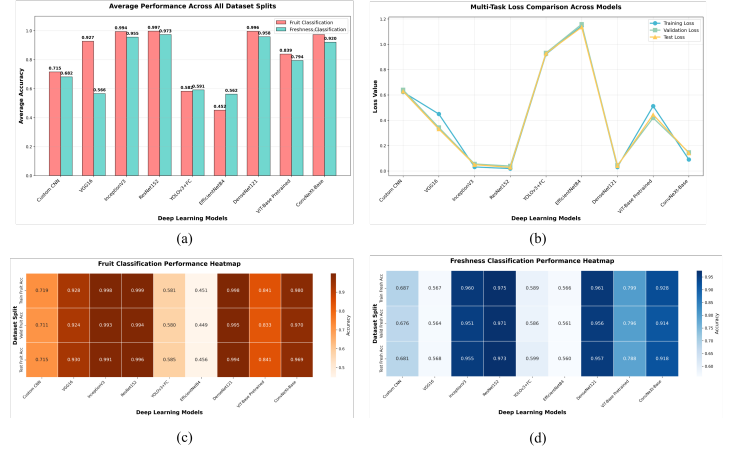


Fig. 2. Comparative performance analysis of nine deep learning models: (a) average accuracy across both tasks, (b) total loss values during training, (c) fruit classification accuracy across train/validation/test splits, and (d) freshness classification accuracy across dataset splits.

freshness probability, facilitating binary cross-entropy training and straightforward thresholding in deployment.

Joint Optimisation: Training minimises a weighted composite loss,

$$L_{\text{total}} = \alpha L_{\text{fruit}} + (1 - \alpha) L_{\text{fresh}}, \quad \alpha = 0.7,$$

balancing the primary classification goal with reliable freshness assessment. This multi-objective formulation serves as implicit regularisation: gradients from freshness detection guide the backbone toward fine-grained surface features that also support better species recognition, leading to the experimentally observed 98.45% combined test accuracy.

III. DEEP LEARNING MODEL COMPARISON

We evaluated nine different deep learning models on our multi-task classification problem: (1) Custom CNN (27M parameters), (2) VGG16 (15.2M parameters), (3) InceptionV3 (23M parameters), (4) ResNet152V2 (59.6M parameters), (5) EfficientNet-B4 (18.8M parameters), (6) DenseNet121 (7.7M parameters), (7) ConvNeXt-Base (88.3M parameters), (8) Vision Transformer (ViT-Base, 27M parameters), and (9) YOLOv3 (8.4M parameters). For transfer learning models, we utilized ImageNet pre-trained weights and fine-tuned them on our dataset using the training procedure described in Section II-D. Fig. 2 presents comprehensive performance comparisons across all nine models. Among the evaluated models, ResNet152V2 and DenseNet121 were the top performers, with ResNet152V2 achieving the highest accuracy of 98.45%. VGG16 performs reasonably for fruit classification but struggles with freshness detection. EfficientNet-B4's poor result (50.81% combined accuracy) is likely due to overfitting given its 18.8M parameters on a relatively small dataset. YOLOv3's low performance (59.18%) is expected since it is designed for object detection rather than fine-grained classification. Our custom CNN, despite having 27M parameters, reached only 69.8% combined accuracy, likely due to limited

TABLE I
CLASSIFICATION REPORT FOR RESNET-152V2 MULTI-TASK
PERFORMANCE

Fruit Classification			Freshness Detection		
Class	Precision	Recall	Class	Precision	Recall
Apple	0.996	0.993	Fresh	0.956	0.992
Banana	0.998	1.000	Rotten	0.992	0.952
Bittergourd	1.000	1.000			
Capsicum	1.000	1.000			
Cucumber	0.995	0.986			
Okra	0.983	0.996			
Orange	0.996	0.993			
Potato	1.000	0.997			
Tomato	0.998	1.000			
Overall Accuracy	99.6%		97.3%		
Macro Average	0.996	0.996	0.974	0.972	
Weighted Average	0.996	0.996	0.973	0.973	

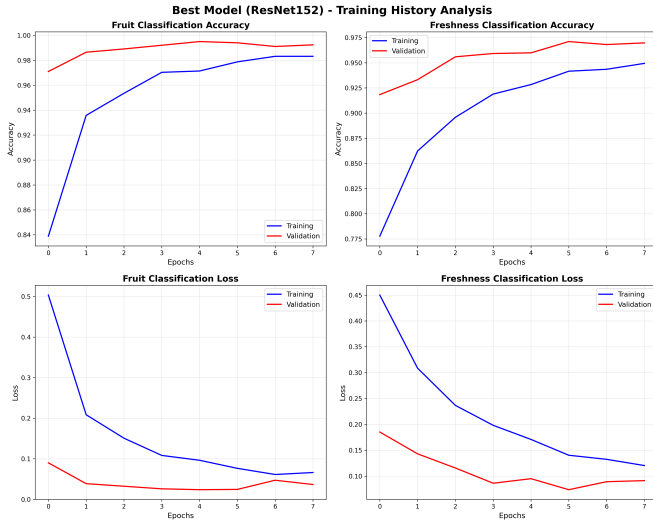


Fig. 3. Training history of ResNet-152V2 showing convergence of accuracy and loss curves for both fruit classification and freshness detection tasks over training epochs.

depth compared to ResNet152V2, reinforcing the advantage of ImageNet-based transfer learning.

IV. PERFORMANCE METRICS OF RESNET-152V2

Based on comparative results, ResNet-152V2 achieved the highest overall performance with 99.64% fruit classification accuracy and 97.27% freshness detection accuracy on the test set, resulting in a combined average accuracy of 98.45% and the lowest test loss of 0.0282. Fig. 3 illustrates the learning curves of ResNet-152V2, demonstrating steady improvement in both training and validation accuracies with no significant gap between them, indicating good generalization. The model achieved rapid convergence within 8-10 epochs with early stopping applied when validation loss ceased to improve. Training was completed in approximately 4.5 hours on Tesla P100 GPU. Table I presents detailed classification metrics, showing excellent precision and recall values across all fruit categories. Most fruit classes achieved perfect or near-perfect

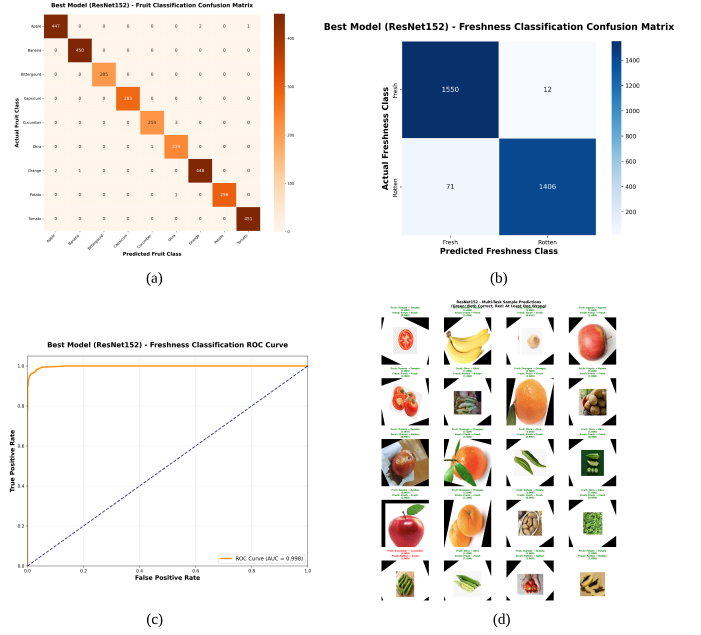


Fig. 4. Performance evaluation of ResNet-152V2 model: (a) Confusion matrix for 9-class fruit type classification, (b) Binary freshness classification confusion matrix, (c) ROC curve for freshness classification, and (d) Sample prediction results showing model confidence scores and visual predictions

performance, with the lowest precision (98.3%) observed for okra classification. Okra's slightly lower precision suggests the model sometimes confuses it with visually similar classes, possibly cucumber or bittergourd. For freshness detection, the model demonstrates strong binary classification performance with 95.6% precision for fresh samples and 99.2% precision for rotten samples. Fig. 4(a) shows the confusion matrices for both tasks. The fruit classification matrix is nearly diagonal with minimal off-diagonal errors, confirming excellent discrimination among all 9 categories. From Fig. 4(b), the binary freshness classification matrix demonstrates high true positive and true negative rates with minimal confusion between fresh and rotten classifications. Fig. 4(c) presents the ROC curve for freshness classification with an exceptional AUC of 0.989, indicating superior discriminative ability between fresh and rotten classes across all threshold values. Fig. 4(d) displays the prediction confidence distribution, revealing that ResNet-152V2 outputs very high confidence for most correct predictions, with incorrect predictions showing appropriately lower confidence values.

V. GRAD-CAM ANALYSIS

To interpret the model's decision-making process, we utilize Gradient-weighted Class Activation Mapping (Grad-CAM) [10] to visualize regions of input images that most influence predictions. Our implementation targets the conv5_block3_out layer (final convolutional layer) with feature map size $7 \times 7 \times 2048$, generating separate heatmaps for each task output. The process involves extracting feature maps from the final convolutional layer, computing gradients

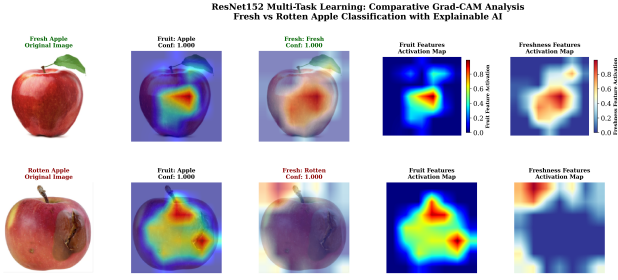


Fig. 5. Grad-CAM visualization for ResNet-152V2 predictions showing fresh (top) and rotten (bottom) apple examples with heatmaps for fruit classification (left) and freshness detection (right). Red regions indicate areas of highest importance for model predictions.

with respect to target class scores for each task, calculating global-average-pooled weights α_k for feature importance, and generating localization maps through weighted combination:

$$L_{\text{Grad-CAM}} = \text{ReLU} \left(\sum_k \alpha_k \cdot A^k \right) \quad (7)$$

where A^k represents the k -th feature map. As shown in Fig. 5, our analysis on fresh and rotten apple examples demonstrates that for fruit classification, the network emphasizes structural characteristics like the apple’s round shape and stem region. For freshness detection, however, attention shifts to surface quality indicators, as fresh apples show uniform surface activation while rotten apples exhibit strong emphasis on brown spots, discoloration, and other surface defects characteristic of spoilage. This validates that our multi-task approach learns task-appropriate features: structural features for fruit type identification and surface quality features for freshness detection.

VI. REAL-WORLD DEPLOYMENT

We deployed our trained ResNet-152V2 model through Hugging Face Spaces [11], creating an interactive web application for real-time fruit freshness classification. The application processes images in approximately 0.3 seconds per inference on CPU, allowing users to upload fruit images and receive instant predictions for both fruit type and freshness status with confidence percentages. The system is publicly accessible at <https://huggingface.co/spaces/nahinfarhan/fruit-classifier> for demonstration purposes. As shown in Fig. 6, the interface demonstrates accurate real-time predictions with high confidence scores for both fresh and rotten banana samples, achieving 100% confidence in correct classifications and providing immediate feedback suitable for practical implementation in markets, storage facilities, and supply chain management systems. This practical validation demonstrates the real-world applicability of our multi-task learning approach and showcases the model’s deployment readiness through modern ML platform infrastructure.

VII. COMPARATIVE ANALYSIS

Table II compares our approach with existing literature. Our multi-task ResNet152V2 model demonstrates superior

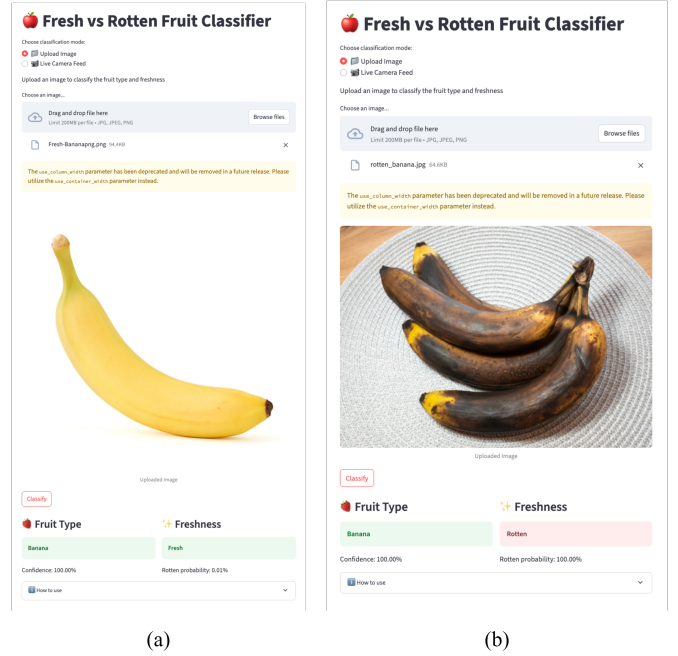


Fig. 6. Deployed web application interface demonstrating real-time fruit freshness classification with high confidence predictions for both (a) fresh banana sample showing 100% confidence in freshness detection, and (b) rotten banana samples with 100% confidence in deterioration state classification.

performance while providing simultaneous fruit identification and freshness assessment. Specifically, our ResNet152V2 model (99.64% fruit, 97.27% freshness) significantly outperforms Zhang et al.’s multi-task CNN (88.66% fruit, 93.24% freshness) by margins of 11% and 4% respectively. While Kang and Gwak achieved slightly higher freshness accuracy (98.50% vs 97.27%), our single model approach exceeds their ensemble’s fruit classification (99.64% vs 97.43%) while requiring fewer computational resources than their dual-model ensemble. The 98.45% combined accuracy significantly outperforms existing multi-task approaches, with our method achieving outstanding fruit classification accuracy compared to other multi-task methods. Our work introduces comprehensive explainability through Grad-CAM visualization (XAI), providing interpretable insights into model decision-making processes that distinguish between structural fruit features and surface quality characteristics, addressing the critical need for transparent AI systems in food quality assessment. Additionally, we demonstrate practical web application deployment using Streamlit and Flask backend with real-time inference capabilities, bridging the gap between research and industrial implementation that most existing studies fail to address.

VIII. CONCLUSION

We developed a comprehensive multi-task deep learning framework for fruit freshness detection that simultaneously classifies fruit type and freshness status. Through extensive benchmarking of nine state-of-the-art deep learning models on the 30,357-image dataset, which is originally imbalanced;

TABLE II
COMPARISON WITH RELATED WORK ON FRUIT FRESHNESS CLASSIFICATION SYSTEMS

Author	Model/Architecture	Dataset tion	Descrip- tion	Task Type	Fruit Acc. (%)	Fresh. Acc. (%)	Deploy.
<i>Multi-Task Learning Approaches</i>							
Proposed Method	ResNet152V2 Multi-task	9 Fruits (30,357 imgs)	Fresh/Rotten Dataset	Fruit + Fresh.	99.64	97.27	Web App
Kang & Gwak [7]	ResNet Ensemble Multi-task	Fresh/Rotten Dataset		Fruit + Fresh.	97.43	98.50	None
Zhang et al. [8]	Multi-task CNN	9 Fruits		Fruit + Fresh.	88.66	93.24	None
Ferrer-Ferrer et al. [12]	Mask R-CNN + Multi-task Head	Fuji Apple, RGB-D images		Fruit + Sizing	85.4	–	None
Afsharpour et al. [13]	Multi-task CNN + Decay	4 Fruits (various decay)		Fresh + ID	95.50	99.93	None
<i>Single-Task Freshness Classification</i>							
Ni et al. [1]	GoogLeNet Transfer	Banana Dataset		Binary Fresh.	–	98.92	None
Pathak & Makwana [2]	6-layer CNN	3 Fruits (10,901 imgs)		Binary Fresh.	–	98.23	None
Ahmed et al. [14]	FreshDNN CNN	8 Fruits (Kaggle, custom)		Binary Fresh.	–	97.80	None
Yuan et al. [5]	CNN-BiLSTM Hybrid	Fruits & Veg.(Mixed)		Binary Fresh.	–	97.76	None
Fahad et al. [6]	VGG-16 + YOLOv5	11 Fruits (60k imgs)		3-class Fresh.	–	82–84	Mobile App

Note: Fresh. = Freshness Classification, Deploy. = Deployment Status, – = not reported

after smart balancing and augmentation, we identified ResNet-152V2 as the optimal architecture, achieving 98.45% combined accuracy (99.64% fruit classification, 97.27% freshness detection) with the lowest test loss of 0.0282. The Grad-CAM analysis reveals that our model learns intuitive feature representations, focusing on structural characteristics for fruit classification and surface defects for freshness detection. This interpretability, combined with the high accuracy and practical deployment, makes our solution immediately applicable to real-world food quality assessment systems. This establishes new state-of-the-art performance for multi-task fruit classification and demonstrates that a single model can effectively handle multiple agricultural classification tasks simultaneously. Future research will focus on expanding the dataset to encompass a broader range of fruit varieties and incorporating multi-level freshness gradations that move beyond binary classification. This will enable the development of more efficient models capable of distinguishing subtle variations in quality, thereby enhancing the applicability of freshness assessment systems in real-world scenarios.

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